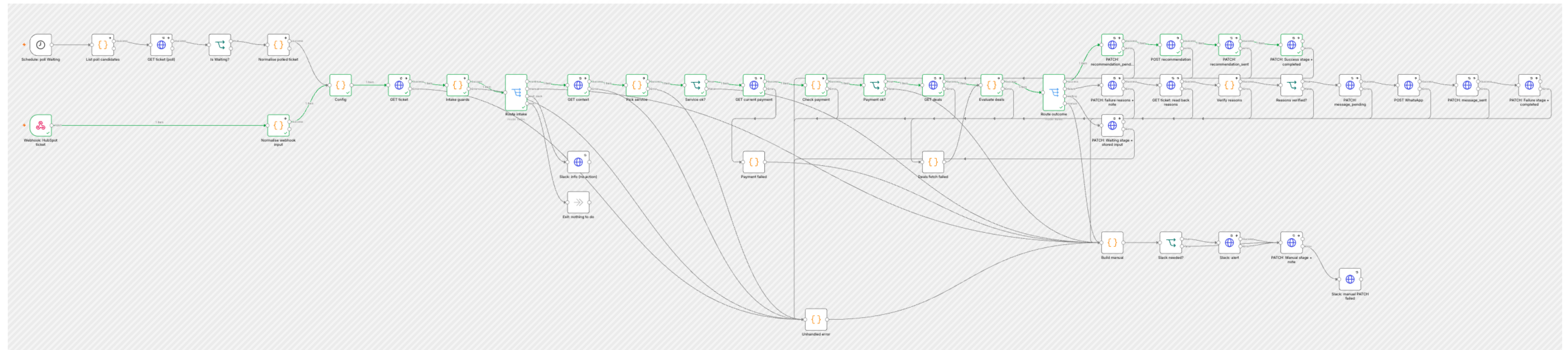


TICKET-001: end to end

Nous MSO mobile recommendation: an n8n workflow that takes a HubSpot ticket in and returns one of four outcomes. These are real executions from the evaluated set (TICKET-001 to TICKET-011), run 17 September 2026.



TICKET-001, end to end. The four outcome branches are all on the canvas; one ticket lights one of them. Status markers bracket the customer-facing POST so an interrupted run is legible.

The eleven evaluated tickets, one run each, plus the scheduled poll that completed 009. Twelve executions, no errors. Outcomes in test/results.md.

TICKET-002: the iD bar, visible in the ranking

Logs

Success in 3.581s

Webhook: HubSpot ticket

Normalise webhook input

Config

GET ticket

Intake guards

Route intake

GET context

Pick service

Service ok?

GET current payment

Check payment

Payment ok?

GET deals

Evaluate deals

Route outcome

PATCH: recommendation_pending

POST recommendation

PATCH: recommendation_sent

PATCH: Success stage + completed

{} Evaluate deals

Success in 9ms

OUTPUT

Success Branch (1 item)Error Branch

```
"evaluation": {
  "feedCount": 3,
  "removed": {
    "malformed": 0,
    "nonApproved": 0,
    "currentProvider": 0,
    "coverage": 0,
    "data": 0
  },
},
"poolCount": 3,
"pool": [
  {
    "stickeeDealId": "SDL-TALK-002",
    "providerId": 9,
    "providerName": "Talkmobile",
    "price": 12.5,
    "monthlyData": 15,
    "coverage": "LIKELY",
    "rankPence": 1250
  },
  {
    "stickeeDealId": "SDL-GIFF-002",
    "providerId": 13,
    "providerName": "Giffgaff",
    "price": 14,
    "monthlyData": 15,
    "coverage": "LIKELY",
    "rankPence": 1400
  },
  {
    "stickeeDealId": "SDL-ID-002",
    "providerId": 18,
    "providerName": "iD Mobile",
    "price": 12,
    "monthlyData": 15,
    "coverage": "LIKELY",
    "rankPence": 1400
  }
]
```

TICKET-002. iD Mobile is 50p cheaper and still loses: rankPence carries a £2 penalty from Config, which changes the ordering and never the saving. The ops Slack thread about iD is a config value, not a special case in the logic.

TICKET-002: the iD bar, visible in the ranking

Logs

Success in 3.581s

Webhook: HubSpot ticket

Normalise webhook input

Config

GET ticket

Intake guards

Route intake

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Pick service

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Payment ok?

GET deals

Evaluate deals

Route outcome

PATCH: recommendation_pending

POST recommendation

PATCH: recommendation_sent

PATCH: Success stage + completed

{} Evaluate deals

Success in 9ms

OUTPUT

Success Branch (1 item)

Error Branch

"coverage": "LIKELY",
"rankPence": 1250

},
{
"stickeeDealId": "SDL-GIFF-002",
"providerId": 13,
"providerName": "Giffgaff",
"price": 14,
"monthlyData": 15,
"coverage": "LIKELY",
"rankPence": 1400

},
{
"stickeeDealId": "SDL-ID-002",
"providerId": 18,
"providerName": "iD Mobile",
"price": 12,
"monthlyData": 15,
"coverage": "LIKELY",
"rankPence": 1400

}

],
"best": {
"stickeeDealId": "SDL-TALK-002",
"providerId": 9,
"providerName": "Talkmobile",
"price": 12.5,
"monthlyData": 15,

TICKET-002. iD Mobile is 50p cheaper and still loses: rankPence carries a £2 penalty from Config, which changes the ordering and never the saving. The ops Slack thread about iD is a config value, not a special case in the logic.

TICKET-002: the iD bar, visible in the ranking

Personal / Nous MSO mobile recommendation v1

Executions

No active executions

Auto refresh

Sep 17, 14:02:25

Succeeded in 6.067s

Sep 17, 14:02:20

Succeeded in 3.581s

Sep 17, 14:02:15

Succeeded in 3.593s

Sep 17, 14:02:10

Succeeded in 4.872s

Sep 17, 14:02:06

Succeeded in 4.807s

Sep 17, 14:02:02

Succeeded in 661ms

Sep 17, 14:02:01

Succeeded in 4.996s

Sep 17, 14:01:56

Succeeded in 5.377s

Sep 17, 14:01:51

Succeeded in 1.728s

Sep 17, 14:01:46

Succeeded in 3.461s

Sep 17, 14:01:26

Succeeded in 3.698s

Sep 17, 14:00:02

Succeeded in 7.851s

Sep 17, 13:58:57

Succeeded in 5.945s

Which executions is this workflow saving?

Editor

Executions

Evaluations

Sep 17, 14:02:20

Succeeded in 3.581s | 70KB | ID#143 | v1 phase 4 test (poll 2 min)

Copy to editor

Logs

Success in 3.581s

Webhook: HubSpot ticket

Normalise webhook input

Config

GET ticket

Intake guards

Route intake

GET context

Pick service

Service ok?

GET current payment

Check payment

Payment ok?

GET deals

Evaluate deals

Route outcome

PATCH: recommendation_pending

POST recommendation

PATCH: recommendation_sent

PATCH: Success stage + completed

Evaluate deals

Success in 9ms

Input

Output

...

OUTPUT

Success Branch (1 item)

Error Branch

```
},
"ticket": {
  "id": "TICKET-002",
  "subject": "MOB | Alex Patel | Make recommendation | Mobile #2",
  "hs_pipeline_id": "228462820",
  "hs_pipeline_stage": "390658766",
  "automation_status": null,
  "next_task_description": null,
  "email_switch_failure_reason": null,
  "reporting_switch_failure_reason": null
},
"serviceIndex": 2,
"firstName": "Alex",
"attempts": 0,
"decision": {
  "route": "success",
  "shape": "S1",
  "cause": null,
  "note": "Recommending Talkmobile 15GB £12.5/mo, saves £7.50/mo. stickeeDealId: SDL-TALK-002. 3 deals in feed, removed 0 malformed, 0 non-approved, 0 current provider, 0 coverage, 0 data. [webhook]",
  "slack": null
},
},
"context": {
  "id": "HH-002",
  "postcode": "M1 4WP",
  "userId": "USR-002",
  "serviceCount": 2
},
"service": {
  "id": "MOB-002B",
  "createdAt": "2024-03-20T12:00:00Z",
```

TICKET-002. iD Mobile is 50p cheaper and still loses: rankPence carries a £2 penalty from Config, which changes the ordering and never the saving. The ops Slack thread about iD is a config value, not a special case in the logic.

TICKET-004: both reasons written before the stage moves

Logs

Success in 4.807s

Route intake

GET context

Pick service

Service ok?

GET current payment

Check payment

Payment ok?

GET deals

Evaluate deals

Route outcome

PATCH: failure reasons + note

GET ticket: read back reasons

Verify reasons

Reasons verified?

PATCH: message_pending

POST WhatsApp

PATCH: message_sent

PATCH: Failure stage + completed

POST WhatsApp Success in 424ms

INPUT

1 item

TidTICKET-004

properties

TidTICKET-004

TsubjectMOB | Casey Wright | Make recommendation | Mobile #1

Ths_pipeline_id228462820

Ths_pipeline_stage390658766

Tautomation_statusmessage_pending

Tnext_task_descriptionBest eligible: none (cause: coverage). 3 deals in feed, removed 0 malformed, 0 non-approved, 0 current provider, 3 coverage, 0 data. [webhook]

Temail_switch_failure_reasonMob - can't beat deal

Treporting_switch_failure_reasonINELIGIBLE_COULD_NOT_BEAT_CURRENT_DEAL

OUTPUT

1 item

Success Branch (1 item)

Error Branch

[{"messageId": "MSG-1789650129754-c2funp", "status": "sent"}]

The brief's post-mortem was a ticket moved with only one reason set. Here both are written in one PATCH, read back from the API and checked against the allowed lists before anything else happens. The stage is still 390658766 at this point; it moves last, after the message is confirmed sent. A read-back that disagrees routes to Manual instead.

TICKET-004: the customer-facing judgement

Logs

Success in 4.807s

Webhook: HubSpot ticket

Normalise webhook input

Config

GET ticket

Intake guards

Route intake

GET context

Pick service

Service ok?

GET current payment

Check payment

Payment ok?

GET deals

Evaluate deals

Route outcome

PATCH: failure reasons + note

GET ticket: read back reasons

Verify reasons

{} Evaluate deals

Success in 16ms

Input

Output

...

▼

OUTPUT

Success Branch (1 item)

Error Branch

"nonApproved": 0,

"currentProvider": 0,

"coverage": 3,

"data": 0

}

,

"poolCount": 0,

"pool": [

,

"best": null,

"shadow": null,

"fetchError": null,

"savingPence": null

,

"whatsappMessage": "Hi Casey, it's Nous.\nWe've just looked at the SIM-only deals available at your address.\nThe deals we found aren't expected to have good coverage at your postcode, so we're not going to move you onto something that might not work where you live.\nIf you'd like us to look again, or you've spotted a deal you'd like us to check, just reply here.",

"failurePatch": {

"next_task_description": "Best eligible: none (cause: coverage). 3 deals in feed, removed 0 malformed, 0 non-approved, 0 current provider, 3 coverage, 0 data. [webhook]",

"email_switch_failure_reason": "Mob - can't beat deal",

"reporting_switch_failure_reason": "INELIGIBLE_COULD_NOT_BEAT_CURRENT_DEAL"

}

}

]

1 item

Copy

004 failed on coverage, not price, so the customer is told about coverage. Three templates keyed on failureCause. The reporting enum has no coverage value, so the ticket says INELIGIBLE_COULD_NOT_BEAT_CURRENT_DEAL and the customer gets the truth. That gap is my pushback item.

TICKET-009: inventory refreshes through the day

Logs

Success in 5.945s

- Webhook: HubSpot ticket
- Normalise webhook input
- Config
- GET ticket
- Intake guards
- Route intake
- GET context
- Pick service
- Service ok?
- GET current payment
- Check payment
- Payment ok?
- GET deals
- Evaluate deals
- Route outcome
- PATCH: Waiting stage + stored input

⊕ PATCH: Waiting stage + stored input

Success in 574ms

Input

Output

...

⌵

OUTPUT

Success Branch (1 item)

Error Branch

Q

⌵

⌶

{ }

1 item

📄

```
[
  {
    "id": "TICKET-009",
    "properties": {
      "id": "TICKET-009",
      "subject": "MOB | Charlie Nguyen | Make recommendation | Mobile #1",
      "hs_pipeline_id": "228462820",
      "hs_pipeline_stage": "5060559097",
      "automation_status": "waiting",
      "next_task_description": "Attempt 1 of 3: no eligible deals in feed. [webhook]",
      "email_switch_failure_reason": null,
      "reporting_switch_failure_reason": null,
      "automation_attempts": 1,
      "automation_household_id": "HH-009",
      "automation_user_id": "USR-009",
      "automation_test_mode": true
    }
  }
]
```

009's first deals call came back empty. Rather than failing the ticket, the run parks it at Waiting with everything the next attempt needs written onto the ticket itself, and a scheduled poll picks it up. No n8n Wait node, so nothing is held in memory and a restart loses nothing. The note ends [schedule] so you can tell from the ticket which path completed it.

TICKET-009: inventory refreshes through the day

Logs

Success in 7.851s

Schedule: poll Waiting

List poll candidates

GET ticket (poll)

Is Waiting?

Normalise polled ticket

Config

GET ticket

Intake guards

Route intake

GET context

Pick service

Service ok?

GET current payment

Check payment

Payment ok?

GET deals

Evaluate deals

Route outcome

PATCH: recommendation_pending

POST recommendation

PATCH: recommendation_sent

PATCH: Success stage + completed

Schedule: poll Waiting

Success in 0s

...

OUTPUT

[

{

"timestamp": "2026-09-17T14:00:02.000+01:00",

"Readable date": "September 17th 2026, 2:00:02 pm",

"Readable time": "2:00:02 pm",

"Day of week": "Thursday",

"Year": "2026",

"Month": "September",

"Day of month": "17",

"Hour": "14",

"Minute": "00",

"Second": "02",

"Timezone": "Europe/London (UTC+01:00)"

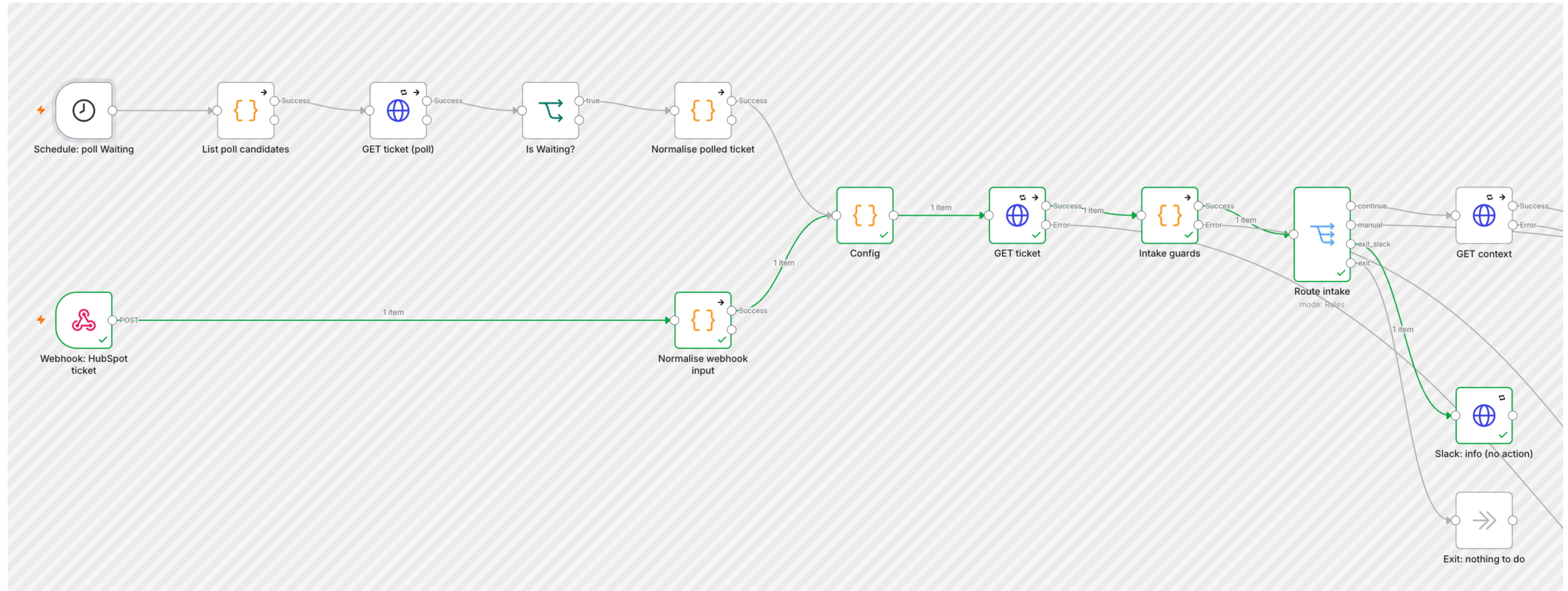
}

]

1 iter

009's first deals call came back empty. Rather than failing the ticket, the run parks it at Waiting with everything the next attempt needs written onto the ticket itself, and a scheduled poll picks it up. No n8n Wait node, so nothing is held in memory and a restart loses nothing. The note ends [schedule] so you can tell from the ticket which path completed it.

TICKET-010: safe to re-run



010 arrives with automation_status: completed already set. The run re-reads the ticket from the API rather than trusting the webhook, recognises it, posts one informational Slack line because a completed marker on an open stage is worth a human knowing about, and stops. Zero writes. Every evaluated ticket is now in this state, so every one of them is re-runnable without damage.

TICKET-006: a transient upstream failure, absorbed

The screenshot displays the n8n workflow editor for a node named "GET deals". The interface is divided into three main sections: INPUT, SETTINGS, and OUTPUT.

INPUT: The node is configured with a "Payment ok?" trigger. The JSON input data is as follows:

```
[
  {
    "ticketId": "TICKET-001",
    "householdId": "HH-001",
    "userId": "USR-001",
    "testMode": true,
    "source": "webhook",
    "cfg": {
      "baseUrl": "https://nous-case-study-saffron.vercel.app",
      "SAVING_FLOOR_PENCE": 300,
      "ID_PENALTY_PENCE": 200,
      "APPROVED_PROVIDER_IDS": [
        9,
        13,
        18
      ],
      "ID_MOBILE_PROVIDER_ID": 18,
      "COVERAGE_ACCEPTED": [
        "LIKELY"
      ],
      "MAX_WAITING_ATTEMPTS": 3,
      "PIPELINE_ID": "228462820",
      "stages": {
        "open": "390658766",
        "success": "446512118",
        "failure": "409734350",
        "manual": "440806104",
        "waiting": "5060559097"
      },
      "SUBJECT_REGEX": "^MOB \\| (.+?) \\| Make recommendation \\| Mobile #(?\\d+)",
      "NO_SERVICE_MARKER": "#No service found",
      "ATTEMPTS_PROPERTY": "automation_attempts",
      "INTERRUPTED_STATUSES": [
        "recommendation_pending",
        "recommendation_sent",
        "message_pending",
        "message_sent"
      ],
      "EMAIL_REASONS": [

```

SETTINGS: The "Settings" tab is active. The configuration includes:

- SSL Certificates: ☐
- Always Output Data: ☐
- Execute Once: ☐
- Retry On Fail: ☒
- Max. Tries: 4
- Wait Between Tries (ms): 2,000
- On Error: Continue (using error output)
- Notes: (Empty text area)
- Display Note in Flow?: ☐

HTTP Request node version 4.2 (Latest version: 4.5)

OUTPUT: The "Success Branch (1 item)" is active. The JSON output data is as follows:

```
[
  {
    "deals": [
      {
        "stickeeDealId": "SDL-GIFF-001",
        "providerName": "Giffgaff",
        "providerId": 13,
        "monthlyData": 30,
        "effective_line_rental": 15,
        "coverageAtHouseholdLocation": "LIKELY"
      },
      {
        "stickeeDealId": "SDL-TALK-001",
        "providerName": "Talkmobile",
        "providerId": 9,
        "monthlyData": 25,
        "effective_line_rental": 18,
        "coverageAtHouseholdLocation": "LIKELY"
      },
      {
        "stickeeDealId": "SDL-ID-001",
        "providerName": "iD Mobile",
        "providerId": 18,
        "monthlyData": 30,
        "effective_line_rental": 16,
        "coverageAtHouseholdLocation": "LIKELY"
      },
      {
        "stickeeDealId": "SDL-VOXI-001",
        "providerName": "VOXI",
        "providerId": 7,
        "monthlyData": 30,
        "effective_line_rental": 12,
        "coverageAtHouseholdLocation": "LIKELY"
      }
    ]
  }
]
```

GET deals retries before it fails: four tries, two seconds apart, then falls through to the error branch rather than surfacing a 500 to the ticket. This is the policy that absorbed 006's first-attempt 500 on the day; the ticket shows a clean Success with no sign anything went wrong, confirmed in test/results.md rather than a visible second try here. GETs retry; POST recommendation and POST WhatsApp do not, because a duplicate there reaches the customer.